# CST521 - Principles of Programming Languages

## CST 521 (Principles of Programming Languages) Syllabus

### Course Information

#### Course Details

- **Course Title:** Principles of Programming Languages
- **Course Number:** CST 521
- **Meeting Location:** Physical
- **Units:** 2

#### Course Description

This course provides a focused study of the fundamental principles behind programming language design and implementation. Key topics include syntax, semantics, and type systems, with an introduction to control structures. Students will explore major programming paradigms, such as procedural and functional programming, gaining an understanding of their practical applications and trade-offs.

#### Materials

- Course textbook: "Programming Language Pragmatics" by Michael L. Scott

#### Technology Requirements

Students must have access to: - A computer (not "chromebook" or similar) with reliable internet connection - Access to Canvas learning management system

#### Course Objectives

At the end of this class, you should be able to:

1. Analyze and compare syntax, semantics, and type systems across different programming languages
2. Understand and implement control structures and their effects on program flow
3. Compare and contrast procedural and functional programming paradigms
4. Evaluate trade-offs in programming language design decisions

### Grading

There are three groups of assignments, briefly described in the table below. **Getting below a 40% in any of these three groups will result in failing the class.**

| Assignment Kind | Value |
|---|---|
| Projects | 35% |
| Exams | 35% |
| Participation | 30% |

### Projects

Projects will take the form of comparative programming exercises across different language paradigms, implementation of simple language features, and analysis assignments examining syntax and semantics. Students will explore fundamental programming language concepts through focused coding and analytical work.

### Exams

Exams will cover material from class and from any assigned reading, as well as draw from knowledge gained during projects.

### Participation

Class is expected to be highly interactive, with students actively answering and asking questions, as well as participating in discussion groups. Therefore, attendance is required, as well as participation in all in-class activities. These activities may take the form of in-class discussions, brainstorming sessions, stand-ups, as well as making active progress on projects during class time.

### Grade Assignment

Grades as assigned based on the standard ranges, which are outlined below.

| Grade Range | Letter Grade |
|---|---|
| $[97, \infty)$ | A+ |
| $[93, 97)$ | A |
| $[90, 93)$ | A- |
| $[87, 90)$ | B+ |
| $[83, 87)$ | B |
| $[80, 83)$ | B- |
| $[77, 80)$ | C+ |
| $[73, 77)$ | C |
| $[70, 73)$ | C- |
| $[60, 70)$ | D |
| $[0, 60)$ | F |

## Syllabus Change Policy

This syllabus is subject to change at the instructor's discretion to enhance learning or accommodate unforeseen circumstances. Students will be notified of any substantive changes (such as changes to due dates, point values of assignments, or major policy modifications) via Canvas announcement and/or email at least one week in advance when possible.

The process for syllabus changes: 1. Instructor identifies need for change 2. Revised syllabus section is prepared 3. Students are notified via Canvas and email 4. Change takes effect after notification period

Students are responsible for staying current with any announced changes to the syllabus.